

components of Apache Spark.

Spark Core

Spark Core is the base engine responsible for large scale parallel and distributed data processing. It handles memory management, scheduling, distributing and monitoring jobs on a cluster, interacting with storage systems and fault recovery. Spark Core is embedded with a fault tolerant, distributed collection of objects that can be operated on in parallel, known as Resilient Distributed Datasets (RDDs).

Spark SQL

Used for structured and semi-structured data processing, Spark SQL is a framework for performing RDBMS operations on data.

Spark Streaming

Spark Streaming is a tool that provides secure and reliable processing of live streaming data. This lightweight framework is ideal for batch processing and real-time streaming of data. It helps convert the incoming live-stream of data into small batches, which are processed by the Spark Engine. The Spark Engine then releases batches of processed data.

Spark MLlib

MLlib is what makes Apache Spark a uniquely useful framework. It is a low level machine learning library of Spark that is scalable and compatible with multiple programming languages. MLlib is capable of performing various machine learning algorithms including clustering, classification, regression and collaborative filtering.

Spark GraphX

GraphX is a well networked graph computation engine of Spark. It can be used for interactive graph computations, exploratory data analysis and is also a great ETL (extract, transform, load) tool.

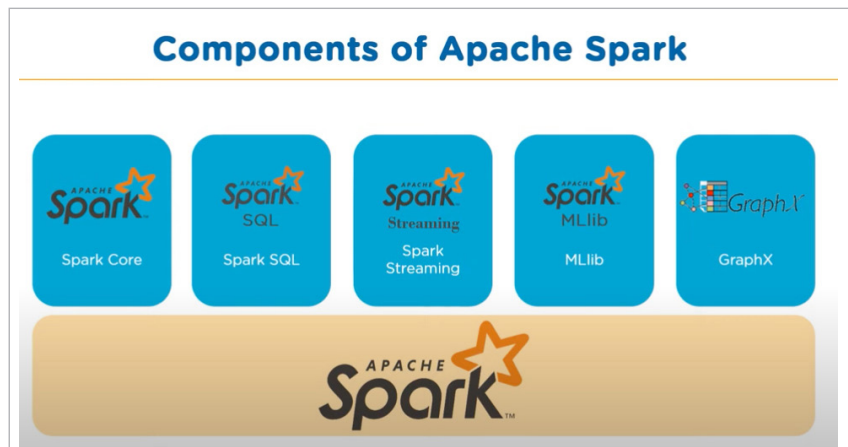


Figure 5: Major components of Apache Spark (Courtesy: Simplilearn)

Kafka

Originally developed by LinkedIn, Apache Kafka is a framework implementation of a software bus using stream processing. If you break down an information transaction on the Internet, it's essentially nothing but an exchange of data between a source computer and a target computer. Each transaction can be thought of as an integration. So if you have one source computer and one target computer, each exchange is a single transaction. But what if you have four source computers and six target computers? Now we need to write 24 integrations and that too with the added complications of protocol selection, data format and schema.

In order to simplify this process, the Kafka framework acts as a broker between the producers (source computers) and consumers (target computers), as shown in Figure 6. Companies such as Airbnb, Netflix, LinkedIn and Walmart, all use Kafka for load balancing and handling requests. Apache Kafka has extremely low latency with the ability to scale to millions of messages per second. It can also be easily integrated with Hadoop and Spark.

Broker

A broker is a Kafka server that runs in a Kafka cluster. Each broker divides a unique transaction between the producer and consumer into a topic. Therefore,

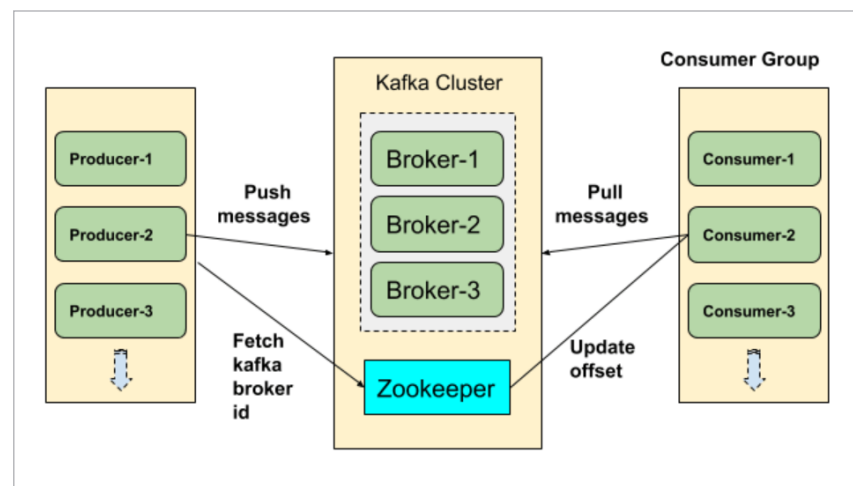


Figure 6: Apache Kafka architecture (Courtesy: Programmer Today)